

AWS Landing Zone Discovery Assessment

1. Executive Summary [🔗](#)

This document outlines the scope, findings, and business value of an AWS Landing Zone (LZ) discovery assessment for the current Kaizen AWS estate. It acts as the foundational diagnostic to guide a structured and risk-aware migration to AWS Control Tower (CT).

2. Purpose [🔗](#)

To assess the current state of AWS accounts across governance, security, networking, and cost posture, and ensure alignment with AWS Control Tower prerequisites and capabilities.

3. Objectives [🔗](#)

- Map all AWS accounts and core configurations in the estate.
- Assess Control Tower readiness by evaluating foundational services.
- Identify compliance, security, and automation gaps.
- Segment accounts by readiness tier for phased migration.
- Provide audit trails and actionable recommendations.

4. Discovery Approach [🔗](#)

A centralized Python-based tool was executed from the audit/management account to:

- Assume roles into all organization accounts via `sts:AssumeRole`.
- Perform read-only, API-based discovery using AWS SDK (`boto3`).
- Capture service posture and account-level metadata across 10+ domains.
- Generate version-controlled CSV output for downstream reporting.

5. Discovery Scope & Capabilities [🔗](#)

The script captures detailed metadata in the following categories:

Category	Description
Account Inventory	Active accounts, OUs, creation dates.
IAM & Access Mgmt	IAM roles, attached policies, boundary policies, unused entities.
Security Tools	GuardDuty, Macie, Security Hub, Inspector statuses.

Audit & Compliance	AWS Config rules, CloudTrail trails, Budgets, and logging posture.
Governance	SCPs, Tag Policies, and Resource Policies (e.g., S3 bucket policies).
Networking	VPCs, subnets, NATs, TGWs, and Network Firewall usage.
CT Readiness	Enabled guardrails, Control Tower setup per account, compatibility markers.
Cost & Usage	Lambda/Glue/Athena/S3 usage, regional spend, tag coverage, idle flags.
Automation Inventory	Lambda, Step Functions, EventBridge rules, scheduled workflows.
Service Catalog & AFC	Detection of AFC pipelines and Service Catalog products.

The discovery script uses modular, role-assumed sessions to perform a read-only assessment across accounts. The output is written to a downloadable CSV for ongoing analysis and reporting.

6. Sample Output Fields [🔗](#)

Example CSV includes:

- `AccountId` , `AccountName` , `Region` , `IAMRoles` , `ConfigEnabled` , `Budgets` , `SCPCount`
- `CTGuardrailsEnabled` , `MacieStatus` , `LambdaCount` , `GlueJobCount` , `TopServices`
- `TaggedResourcesPercent` , `S3PublicBuckets` , `UnencryptedBuckets` , `Estimated30dCostUSD`

Enables filtering and prioritization by readiness or risk.

7. Script Enhancements & Appendages [🔗](#)

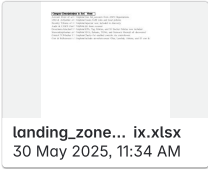
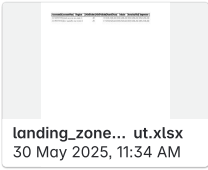
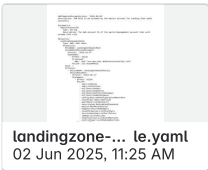
Additional modules were appended to the base script to provide:

- Tag Policy Compliance Engine for enforcement-ready tagging checks.
- Guardrail Compatibility Checker to anticipate policy conflicts.
- Budget Discovery & Alert Readiness to flag missing cost controls.
- S3 Policy Validator to check for overly permissive or missing policies.
- Lambda/Glue/Athena Usage Tracker for serverless cost posture.

Output destinations:

- Versioned S3 folders, Athena query-ready.
- Dashboard-ready output for ingestion into QuickSight or Excel.

8. Key Artifacts [🔗](#)

Artifact	Description	Purpose	Attachment
Coverage Matrix	Comprehensive matrix of all implemented discovery capabilities with feature completeness indicators	Reflects all implemented discovery capabilities and their status (Complete, In Progress, Planned)	 landing_zone... ix.xlsx 30 May 2025, 11:34 AM
Landing Zone Discovery Output	Sample CSV/JSON output per account and region with detailed configuration findings	Provides raw data for analysis, reporting, and planning migration waves	 landing_zone... ut.xlsx 30 May 2025, 11:34 AM
Landing Zone Discovery Script	Python-based discovery script with documentation and execution instructions	Enables repeatable, consistent discovery process that can be executed by the team	 landing_zone... very.py 02 Jun 2025, 11:01 AM
Landing Zone Audit IAM role	IAM role deployed to all AWS Organization member accounts (and the management account) to allow secure, read-only access for centralized Landing Zone configuration discovery and readiness assessment.	Enables a discovery script or automation pipeline (run from the management account) to assume into each member account and collect metadata about IAM, networking, audit, and security service configurations. The collected data supports readiness scoring, configuration drift reporting, and dashboarding.	 landingzone-... le.yaml 02 Jun 2025, 11:25 AM
Readiness Assessment Dashboard	Interactive QuickSight/Excel dashboard that	Allows stakeholders to quickly assess migration readiness	

	visualizes account migration readiness with multiple filters (e.g., by OU, by service enablement status, by compliance level). Features heat maps, readiness scores (0-100%), and drill-down capabilities for detailed analysis.	across the organization, identify patterns, and prioritize remediation efforts. Provides executive-friendly visualizations for steering committee reviews.	
Configuration Drift Report	Detailed comparison report that maps each account's current configuration against Control Tower's desired state baseline. Identifies specific deltas in IAM permissions, SCPs, guardrails, and network configurations. Categorizes findings by severity (Critical, High, Medium, Low).	Identifies precise configuration gaps that need remediation before migration. Serves as input for remediation playbooks and helps estimate remediation effort. Provides traceability for compliance and audit purposes.	

9. Business Value [🔗](#)

Benefit	Value Proposition
Visibility	Transparent account configuration baseline.
Risk Mitigation	Early detection of misconfigurations and non-compliance.
Migration Velocity	Informs wave planning and reduces rework.

Cost Control	Surfaces opportunities for tagging, budgeting, and right-sizing.
Security Readiness	Confirms if foundational services are enabled across the estate.
Stakeholder Alignment	Creates a common language across Devops, Security, and Finance.

10. Next Steps [🔗](#)

- Disseminate discovery report to core stakeholders (Ops, Sec, FinOps).
- Tier accounts by readiness (High, Medium, Low).
- Plan remediation playbooks per tier (e.g., SCP fixes, tagging enforcement).
- Design phased migration waves (Wave 0 = PoC, Wave 1 = Low-risk, etc.).
- Launch executive steering committee and success tracking dashboard.
- Finalize Control Tower configuration baseline and OU structure.

11. Ownership & Execution [🔗](#)

- Discovery Owner: [DevOps / Governance Team]
- Script Repository: [Internal Git or version-controlled repo]
- Execution Cadence: [Initial + pre-migration wave re-runs]
- Output Storage: [S3 with restricted access or secure workspace]
- Stakeholder Distribution: [Email groups, dashboards, and project trackers]

12. Proposed Timeline [🔗](#)

Milestone	Target Date
Discovery Launch	[Start Date]
Report Completion	[Report Date]
Readiness Tiering Complete	[Tiering Date]
Wave 0 Migration	[Pilot Date]
Full Migration Go-Live	[Target Q/Year]

13. References & Resources [🔗](#)

Resource	Description	URL
AWS Control Tower Documentation	Official AWS documentation for Control Tower	https://docs.aws.amazon.com/controltower/

AWS Control Tower Implementation Guide	Step-by-step implementation guidance	Getting started with AWS Control Tower - AWS Control Tower
AWS Landing Zone to Control Tower Migration	AWS guidance on migration paths	https://aws.amazon.com/blogs/mt/migrating-from-aws-landing-zone-to-aws-control-tower/
AWS Well-Architected Framework	Best practices for architecting in AWS	AWS Well-Architected - Build secure, efficient cloud applications
AWS Config Rules	Reference for AWS Config managed rules	List of AWS Config Managed Rules - AWS Config
AWS Organizations SCPs	Documentation on Service Control Policies	Service control policies (SCPs) - AWS Organizations
AWS Security Hub	Security posture management reference	What is AWS Security Hub CSPM? - AWS Security Hub
AWS Boto3 Documentation	Python SDK reference for building discovery tools	Boto3 1.38.44 documentation
AWS Cloud Operations Blog	Migrate AWS Landing Zone solution to AWS Control Tower	Migrate AWS Landing Zone solution to AWS Control Tower AWS Cloud Operations Blog